Case 43: WU-E ellipse response

1 Verification purpose

This case verifies the coupled response from Wildland–Urban ELMFIRE (WU–E) source wind and building geometry through the model-2 ellipse and heat-flux routines to the transient direct-flame-contact (DFC) and radiation rasters. It is capable of exposing an incorrect wind-unit conversion, a wrong branch predicate, a regression-coefficient or ellipse-construction error, a wind-direction sign or raster-orientation error, failure of the separation clamp, and disagreement between constant and raster building-parameter input paths.

One initially burning structure is held stationary by a zero spread-adjustment raster. Spotting, terrain, moisture variation, and secondary propagation are disabled. Twenty-three deterministic simulation conditions exercise both source-wind branch boundaries, Hamada parameters (A) and (D), four right-angle rotations, and constant/raster equivalence, plus the intended source fuel-reach factors 1 for structures and 1/3 for vegetation. The reference calculation implements the intended downscaled WU–E formulation. The pinned source is the system under test, so a mismatch is reported rather than copied into the expected result.

Verification question. Do all required simulation conditions reproduce the intended heat-release rate (HRR) per unit area (HRRPUA) and full-domain DFC/radiation maps within normalized L_1 error 0.005, preserve the paired clamp, rotation, and constant/raster invariants, and create no additional burning cell?

Current result: PASS. 23 of 23 required simulation conditions have evaluable evidence. Analysis procedure: **COMPLETE.**

2 Mathematical formulation and numerical implementation

The source converts the 20-ft input wind from miles per hour to metres per second as $V = 0.447W$. The regression is linear when $V < 10$, quadratic when $10 \le V \le 17.3$, and linear with a distinct coefficient set when $V > 17.3$. In each branch it produces downwind, upwind, and sidewind reaches (K_d, K_u, K_s) from (V, A, D), after enforcing

$$D_{\text{eff}} = \min(D, 50\,\text{m}). \tag{1}$$

The stored ellipse quantities are

$$B = \frac{K_d + K_u}{2}, \qquad E = \min\left(\frac{B}{2}, B - K_u\right), \qquad A_e = \frac{K_s}{\sqrt{1 - (E/B)^2}}. \tag{2}$$

Here B, A_e, and E retain the source nomenclature major, minor, and eccentricity; all three are lengths. For comparison, the pinned source currently constructs the following continuous angular boundary, which is a documented deviation from the intended three-sector model:

$$r_e(\theta_f) = \frac{0.3K_d(B-E)}{A_e^2}\,\frac{A_e^2}{B - E\cos\theta_f}, \qquad \theta_f = \text{atan2}(\Delta y, \Delta x) - \frac{\pi}{180}(270 - WD). \tag{3}$$

The intended model uses a three-sector reach and a physical-distance correction. With $L_c =$ `BLDG_AREA` interpreted as a linear characteristic building dimension in metres and s_{sd} the separation,

$$\beta_{ssd} = \min\left(1, \frac{s_{sd} + L_c}{\Delta x}\right), \tag{4}$$

$$K_\theta = \begin{cases} B + E, & \cos\theta_f > 0.7, \\ B - E, & \cos\theta_f < -0.7, \\ 2A_e, & \text{otherwise}, \end{cases} \tag{5}$$

and the target-cell overlap factors are clipped to $[0, 1]$:

$$f_d = \text{clip}\left(\frac{K_\theta + \Delta x/2 - r\beta_{ssd}}{\Delta x}, 0, 1\right), \tag{6}$$

$$f_r = \text{clip}\left(\frac{HAZ + \Delta x/2 - r\beta_{ssd}c}{\Delta x}, 0, 1\right)(1 - f_d). \tag{7}$$

With $A_s = L_c^2$, $C_h = A_s/[\pi(cB)(cA_e)]$, default $c = 1$, target nonburnable fraction $n_b = 0$, and absorptivity $\alpha = 0.8$, the expected fields are

$$q''_{\text{DFC}} = g_f \eta(W)\, 0.65(1 - n_b) f_d\, \text{HRRPUA}(t) C_h, \tag{8}$$

$$q''_{rad} = g_f \eta(W) \frac{0.35(1 - n_b)\alpha f_r \text{HRRPUA}(t) A_s}{4\pi(r\beta_{ssd}c)^2}, \tag{9}$$

$$\eta(W) = \frac{3.25}{1 + \exp[-0.05(W - 75)]} + 0.5, \qquad g_f = \min\left[\left(\frac{\Delta x}{L_c}\right)^2, 1\right]. \tag{10}$$

The physical heat-affecting zone is $HAZ = 100$ m and its search radius is $\lceil HAZ/\Delta x \rceil$. Radiation is complementary to DFC through $(1 - f_d)$. The source instead uses a continuous angular ellipse, 0.3 heat factors, cell-area normalization, and a cell-count band; these are explicit deviation targets. Preprocessing assigns the prescribed parameter values independently to each simulation condition.

3 Derivation of expected behavior

The ellipse-regression coefficients and branch predicates were independently transcribed from the specified WU-E spread-rate formulation. The reviewed configuration interpretation, spread-rate calculation, and level-set formulation were identical to the specified ELMFIRE source revision; this identity claim does not extend to all other model components. Purnomo et al. describe the wind-responsive urban flame ellipse and heat-transfer coupling in the peer-reviewed WU–E publications [1, 2]. The detailed equations above document the intended implemented model directly. The executable snapshot is evaluated against, rather than substituted for, this formulation.

At $V = 10$ and $V = 17.3$ m s^{-1}, equality belongs to the quadratic branch. The immediate neighboring simulation conditions on each side must instead use the linear low- or high-wind coefficients. The $D = 75$ m field must be identical to $D = 50$ m. A raster parameter field equal to the constants must produce the same heat maps. With all other inputs invariant, rotating the wind-from direction by $90°$ must rotate the complete DFC-plus- radiation field by the same angle on this square grid.

The HRRPUA curve rises from zero to 100 kW m^{-2} over one second, remains at the peak through $t = 100$ s, decays linearly to zero at $t = 110$ s, and stays zero. The $t = 60$ s dump consequently gives an exact nonzero full-map comparison without conflating ellipse response with the HRR transient.

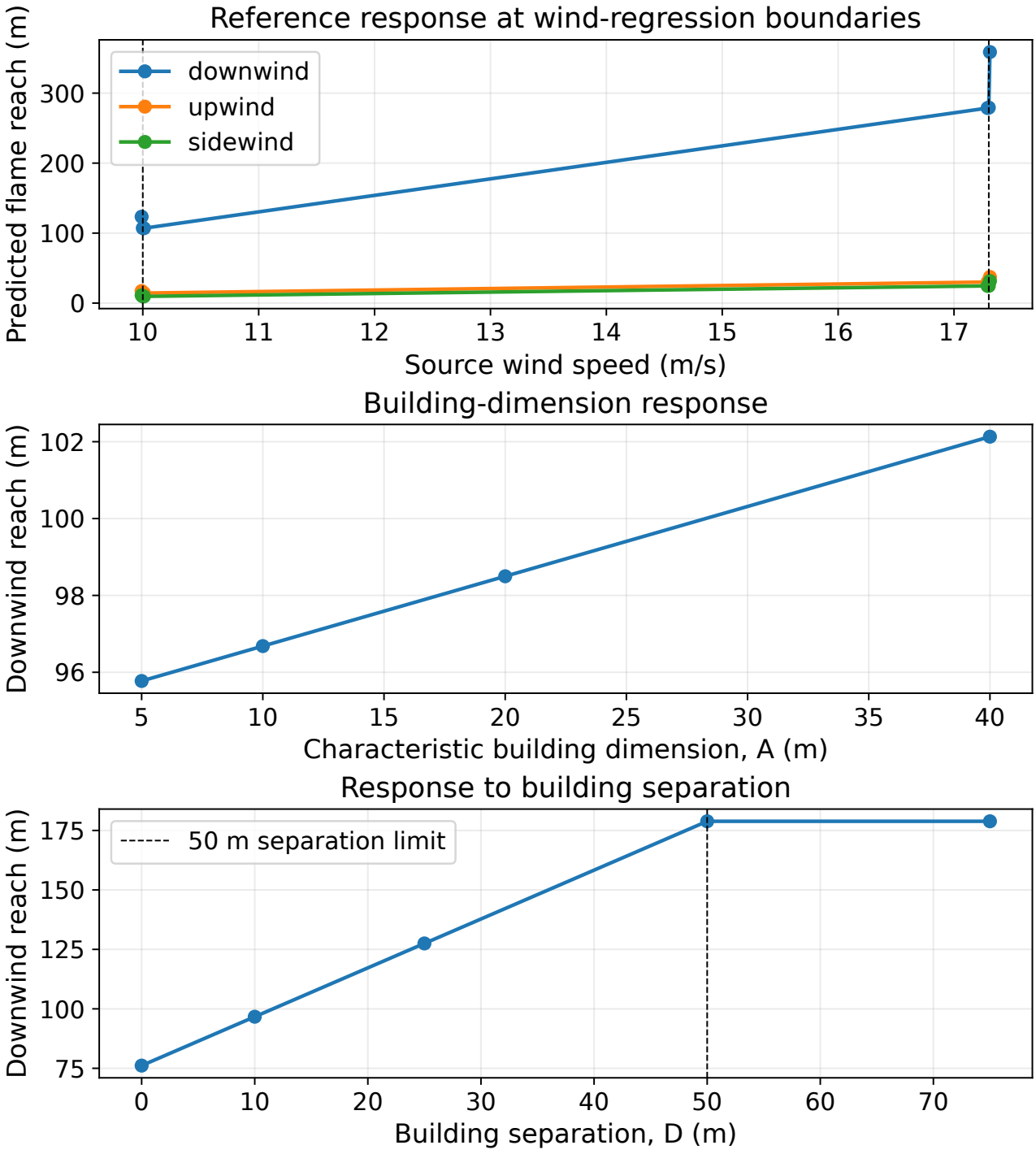


Figure 1: Intended-model reference calculation for the branch, characteristic building dimension, and separation sweeps. These are predicted values, not ELMFIRE output.

4 Verification metrics and acceptance criteria

For any reference raster R and ELMFIRE raster E, the primary metric is

$$\varepsilon_{L_1} = \frac{\sum_{i,j} |E_{ij} - R_{ij}|}{\sum_{i,j} |R_{ij}|}. \tag{11}$$

The sum covers every grid cell, including zeros outside the WUI update band; nodata and malformed rasters are rejected rather than omitted. HRRPUA and both heat fields use the same metric over all dump times and cells, which also detects an unintended secondary source and requires the final transient maps to be zero after ELMFIRE's step-level reset. A 0.5-percent map tolerance permits single- precision raster output while remaining much smaller than any intended branch or parameter response. Paired algebraic invariants use 10^{-5}; rotation uses the primary 0.5-percent tolerance because trigonometric evaluation occurs in single precision. Each simulation condition must have a confirmed successful completion with matching SHA-256 identities for the simulation settings, spatial inputs, both fuel-property tables, reference and analysis methods, ELMFIRE program, simulation record, and all results. Because ELMFIRE's first scheduled dump occurs after one scheduled output interval, the dump simulation specification must contain exactly 60 and 120 seconds, with a unique final record at the configured stop. Output labels alone cannot establish a valid simulation. The model-program identity is retained, but its relationship to the stated source revision must be established separately.

Expected property	Observable and metric	Criterion	Failure sensitivity
Piecewise source curve	All HRR transient rasters; normalized L_1	≤ 0.005	timing or source contamination
Ellipse heat kernel	Full DFC and radiation maps at peak HRRPUA	each ≤ 0.005	branches, coefficients, orientation
D clamp and input paths	Pairwise full-map normalized L_1	each $\leq 10^{-5}$	clamp or I/O divergence
Rotation covariance	Rotated DFC-plus-radiation maps	≤ 0.005	angle, sign, or raster orientation
Vegetation-source factor	Full DFC/radiation maps for the forest-source simulation condition	each ≤ 0.005	unused 1/3 factor or initial-source HRR suppression
No propagation	Final time of arrival (TOA) and off-source HRR cells	zero additional cells	confounding source creation

All 23 simulation conditions and all required fields must be present. Overall PASS requires every metric to pass; missing, duplicate, malformed, stale, or mistimed evidence produces NOT EVALUABLE.

5 Simulation configuration and predicted outcome

Parameter	Configured value	Isolation role
Grid	55 by 55, 20 m cells	full heat-kernel domain
CRS	EPSG:32610	explicit metric coordinates
Time	$\Delta t = 1$ s, stop 120 s, dump 60 s	exact peak-HRR dump
Initial source	one central fire behavior fuel model (FBFM) 91 cell	stationary WU-E source
Background	nonburnable FBFM93, adjustment 0	prevents propagation
Spotting/terrain	disabled/flat	removes confounders
Model selector	WUI model type 2	UMD/UCB implementation
Ellipse adjustment	1.0	intended default effective-flame factor

The generated constant fields and rasters use identical north-up georeferencing, nodata, dimensions, and data types. The point ignition scheduled at 121 s is deliberately after the 120 s stop; it prevents generic stalled-front termination without adding a physical source during the experiment.

The predicted output is a fixed-source HRR history and wind-oriented DFC and radiation fields. Branch-adjacent simulation conditions may jump because the source itself uses distinct regressions; the test checks that behavior rather than imposing continuity absent from the implementation.

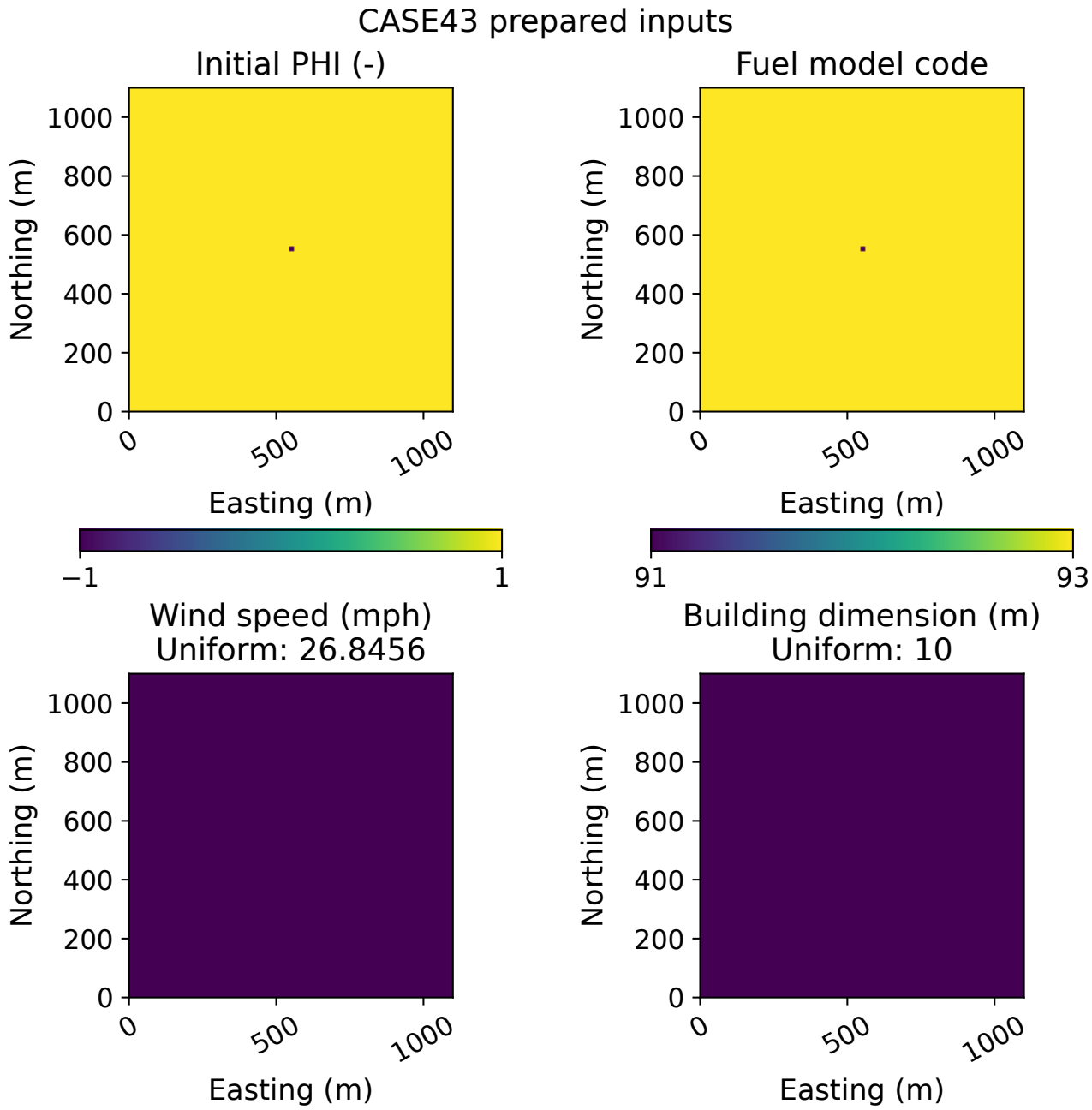


Figure 2: Whole-domain prepared configuration generated from the the 270-degree wind-direction condition GeoTIFF inputs. The single negative PHI cell is the initial WU-E source; FBFM93 and zero adjustment prevent secondary spread.

6 Actual simulation results

The current analysis procedure state is **COMPLETE**; the scientific decision is **PASS**, with 23 of 23 required simulation conditions having evaluable execution evidence. Reason: All required simulation conditions and acceptance metrics passed.. The common executed-binary SHA-256 is `ab47c5e338a3bc3b325df659b0764ce47d5b071e22d07817e4b5cab9f87da672`.

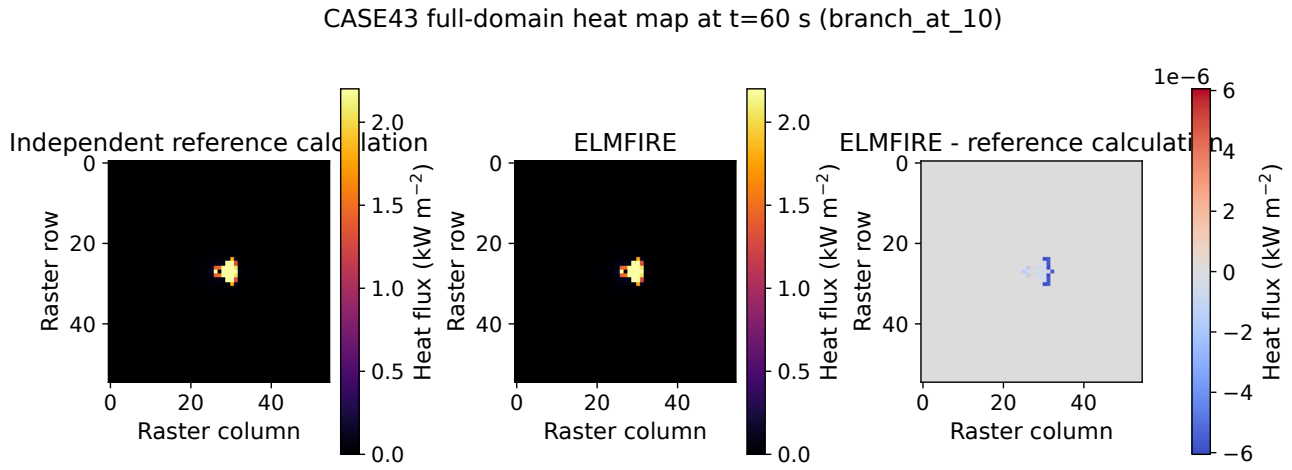


Figure 3: Representative whole-domain ELMFIRE result, independent reference calculation, and difference. This figure is generated only from completed case-specific output.

7 Calculated metrics and verification decision

Maximum HRRPUA normalized L1 error — PASS

Criterion ≤ 0.005

Observed *Units: fraction.*
0

Maximum DFC full-map normalized L1 error — PASS

Criterion ≤ 0.005

Observed *Units: fraction.*
3.02831e-06

Maximum radiation full-map normalized L1 error — PASS

Criterion ≤ 0.005

Observed *Units: fraction.*
1.09758e-06

D greater than 50 m clamp pair error — PASS

Criterion $\leq$ 1e-05

Observed *Units: fraction.*
0

Constant/raster parameter equivalence error — PASS

Criterion $\leq$ 1e-05

Observed *Units: fraction.*
0

Maximum 90-degree covariance error — PASS

Criterion ≤ 0.005

Observed *Units: fraction.*
0

Vegetation-source fuel-factor heat error — PASS

Criterion ≤ 0.005

Observed *Units: fraction.*
1.67242e-06

Maximum arrival-mask mismatch count — PASS

Criterion 0

Observed *Units: cells.*
0

Maximum unexpected HRR source count — PASS

Criterion 0

Observed *Units: cells.*
0

Current decision: PASS. NOT RUN is a analysis procedure state, not evidence of verification and never maps to PASS.

8 Technical interpretation

The current result is limited by missing time-history evidence, rather than a calculated heat-map discrepancy. The solver's low-tagged-node stopping rule terminates this isolated-source setup before the required scheduled heat dumps. The post-stop ignition disables a different, stalled-PHI shortcut; it does not bypass the low-node rule. The lone overrun dump cannot replace the missing history and is not interpreted as a measurement at the configured stop time.

The intended heat kernel also differs from the reviewed source in its angular sectors, heat fractions, wind efficiency, and spatial normalization. These source differences identify what a complete future run should test; they are not numerical verification results from the incomplete run. After the source can sustain a stationary thermal calculation, passing HRR with failing heat maps would isolate the discrepancy to ellipse geometry or heat transfer.

9 Scope and limitations

This case covers the deterministic model-2 source ellipse and transient heat distribution for a stationary structure. It does not verify physical calibration, propagation speed, ignition thresholds, embers, interface transition, fuel consumption, or Statewide WU-E. It intentionally treats the pinned source as the implementation under test and excludes the distinct Statewide WU–E formulation. CASE44 addresses the HRR and heat-parameter response not isolated here.

References

[1] D. M. J. Purnomo et al., "Reconstructing modes of destruction in wildland–urban interface fires using a semi-physical level-set model," *Proceedings of the Combustion Institute*, 40 (2024), 105755. doi:10.1016/j.proci.2024.105755.

[2] D. M. J. Purnomo et al., "Integrating an urban fire model into an operational wildland fire model to simulate one dimensional wildland–urban interface fires: a parametric study," *International Journal of Wildland Fire*, 33 (2024), WF24102. doi:10.1071/WF24102.